

RESEARCH

Open Access



Occlusal parameters and wear of artificial teeth in complete dentures with lingualized versus bilateral balanced occlusion: a randomized clinical trial

Zhiqiang Wang¹, Yan Su², Jianmin Wang¹, Yang Liu¹ and Wenzhong Xing^{3*}

Abstract

Objective This study aimed to assess the occlusal contact area (OCA), occlusal contact number (OCN), bite force, and artificial tooth wear in complete dentures with lingualized and bilateral balanced occlusion.

Methods Edentulous participants who fulfilled the inclusion criteria were divided into the three groups as follows: group I, dentures with lingualized occlusion using interpenetrating polymer network artificial teeth; group II, dentures with lingualized occlusion using double cross-linked artificial teeth; and group III, dentures with bilateral balanced occlusion using anatomical micro-hybrid resin teeth. The silicone techniques were used to assess the OCA and OCN of the mandibular dentures and the Dental Prescale II was used to evaluate the bite force at the 2-week, 3-month, and 12-month follow-up visits. Occlusal contact analyzer software was used to assess the maxillary and mandibular posterior tooth wear at the 12-month follow-up visit. All data was analyzed using the SPSS software.

Results The OCA of group III was significantly higher than that of groups I and II at the 2-week follow-up visit, whereas group I was considerably higher than groups II and III at the 3-month and 12-month follow-up visits. The OCN of the three groups exhibited significant differences at all three follow-up visits. Groups I and II had significantly lower bite force than Group III at the three follow-up visits. At the 12-month follow-up visit, groups I and II had significantly higher wear than Group III.

Conclusions As the duration of use increases, the occlusal contact area of complete dentures increases, regardless of the occlusal schemes. The bite force of lingualized occlusion is typically lower than that of bilateral balanced occlusion. The wear locations of artificial teeth differ between the two types of occlusal schemes. (Retrospectively registered: The Chinese Clinical Trial Registry, No. ChiCTR2300073420(11/07/2023))

Keywords Complete denture, Lingualized occlusion, Occlusal contact area, Bite force, Wear of complete dentures

*Correspondence:

Wenzhong Xing

Xing_wenzhong@163.com

¹Dental Technology Center, Dalian Stomatological Hospital, Dalian 116021, China

²Comprehensive & Emergency Treatment Clinic, Dalian Stomatological Hospital, Dalian 116021, China

³Department of Prosthodontics, Dalian Stomatological Hospital, Dalian 116021, China



© The Author(s) 2024. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Background

Conventional complete dentures were and are still considered to be the most economical and most commonly used treatment option for edentulous participants [1]. They effectively restore function as well as appearance when properly constructed. Adequate retention and stability could be achieved through a good peripheral seal and a stable occlusal relationship [1–3]. Bilateral balanced occlusion (BBO) is an ideal occlusal arrangement for individuals with good alveolar ridge conditions. However, for edentulous participants with compromised ridges, lingualized occlusion (LO) presents several advantages. LO enhances stability by lateral force reduction and concentration of vertical forces on the mandibular residual ridges. This leads to significant improvements in comfort, masticatory efficiency, and quality of life [4, 5].

Primarily, bite force and occlusal contact area (OCA) directly influence the masticatory efficiency of complete dentures [6, 7]. Consequently, the bite force and OCA were consistently the primary focus of the dentists throughout the treatment program [6, 8]. The superior quantitative assessments of occlusal loading and contact area were provided using occlusal contact pressure methods and silicone techniques [9]. Shiga et al. [10], in their study, concluded that the maximum bite force was greater in men than that in women, with the age range of the participants between 20 and 50 years. Imamura et al. [9] found that during a clenching effort at 120% of the maximum voluntary contraction, the average bite force in the molar region on one side measured 347.1 N. Lee et al. [6] concluded that the masticatory performance by silicone techniques increased with the increase in the OCA. Horie et al. [11] observed a correlation between mixing ability and occlusal contact and near contact areas up to a thickness of 200 μm .

The wear of resin artificial teeth affected the clinical performance of complete dentures [12, 13], which might reduce vertical distance and chewing efficiency, leading to masticatory muscle fatigue [14]. The wear of artificial teeth can be evaluated either in vitro or in vivo [14]. In vitro tests are typically two-body or three-body wear tests [12, 13]. Suzuki [15] discovered that polymers with a high degree of cross-linking or inorganic fillers wore out more slowly than conventional polymethyl methacrylate. Heintze et al. [16] reported higher wear in nanohybrid composite resin than that in two different types of ceramics. However, the vitro device could not replicate the complex oral environment, including the relationship of saliva, temperature, food, bite force, and the occlusal cycle on the artificial tooth wear [17–19]. Heintze et al. [20] through the in vivo tests found that the wear increased over time but not linearly. Schmid-Schwab et al. [21] demonstrated that maxilla posterior teeth had significantly greater vertical wear than anterior teeth.

Literature regarding the utilization of digital technology to compare the OCA, OCN, bite force, and the wear of artificial teeth in complete dentures with LO and BBO is limited. This study aimed to assess the OCA, OCN, bite force, and the wear of artificial teeth in complete dentures with LO and BBO. The null hypotheses postulated that there are no differences between LO and BBO regarding the OCA, OCN, and bite force. No differences will be observed in the wear of the three types of artificial teeth after 12 months.

Methods

Study design, IRB approval, and trial registry

A randomized single-blind clinical trial was performed. Participants were randomly allocated into the following three groups: group I, dentures with LO using interpenetrating polymer networks (IPN) artificial teeth (Excellence IPN, Dentsply); group II, dentures with LO using double cross-linked (DCL) teeth (Surpass G, GC), and group III, dentures with BBO using anatomical micro-hybrid resin teeth (Endura, Shofu). The study was approved by the Medical Ethics Committee of Dalian Stomatological Hospital (Ethics Review Batch No. DLKQLL2019015, 28/11/2019), conducted following the Helsinki Declaration, and retrospectively registered in the Chinese Clinical Trial Registry (No. ChiCTR2300073420, 11/07/2023). Informed consent was provided by the participants in the clinical trial.

Participants

Participants with edentulism were admitted to the Department of Prosthodontics of Dalian Stomatological Hospital from October 2019 to December 2021. The edentulous participants who fulfilled the inclusion criteria had a history of complete denture restoration, with their remaining mandibular alveolar ridges classified as II and III [22]. Additionally, symptoms of bruxism, xerostomia, or orofacial motor disorders were not exhibited, nor were there any severe oral manifestations of systematic diseases or psychological/psychiatric conditions that could affect the treatment response. Lastly, the alveolar ridge mucosa had a moderate thickness, and the participants could regularly attend hospital appointments and assist in gathering the necessary data [4, 5].

Sample size

G*Power software with selected statistical analyses of F-test and analysis of variance (ANOVA)-repeated measures within factors were used to calculate the sample size. The parameters used were as follows: input effect size, $f=0.25$; α error=0.05; power=0.80; number of groups=3, and number of measurements=3. The total sample size was 30, resulting in an approximate sample size of 10 per group. The number of participants in each



Fig. 1 Dentures with lingualized occlusion using the interpenetrating polymer network artificial teeth



Fig. 2 Dentures with lingualized occlusion using the double cross-linked artificial teeth

group was increased by one case, considering potential attrition. Thus, a total sample size of 33 participants was required [6, 23].

Randomization

Using the random number table method, the total sample size was divided into three groups. One of the investigators, not involved in evaluating the treatment outcomes, assigned participants to each group based on their order of visit.

Dentures completion

New complete dentures were fabricated based on the following standard procedures: trays and impression compounds (Shangchi, China) were used for recording the preliminary impressions. The edges of the preliminary impressions were modified and used as custom trays to register the final impressions. Border molding was performed using Impregum Penta Soft (3 M ESPE, USA), and the master casts were obtained. The maxillomandibular relationship was recorded using the check bite method, and subsequently, the models were mounted on

an average value articulator (YDM, Japan). The denture teeth were set following the specifications of complete dentures with LO and BBO. After the final try-in, a mandibular closed-mouth impression was made with border molding using the Fit Checker (GC, Japan). All dentures were processed using heat-cured acrylic resin, followed by sanding and polishing. Denture was inserted; Occlusal grinding using articulating paper was performed to detect occlusal errors (Figs. 1, 2 and 3). Before participants left the clinic, they were instructed to visit the clinic for adjustments in time if they felt uncomfortable and were asked to return at 2 weeks, 3 months, and 12 months for follow-up measurements.

Measurement baseline

This study used complete dentures worn for 2 weeks as a measurement baseline to analyze the occlusal parameters and wear of the artificial teeth, as clinical adjustments for pain or discomfort had already been conducted.



Fig. 3 Dentures with bilateral balanced occlusion using the anatomical micro-hybrid resin teeth

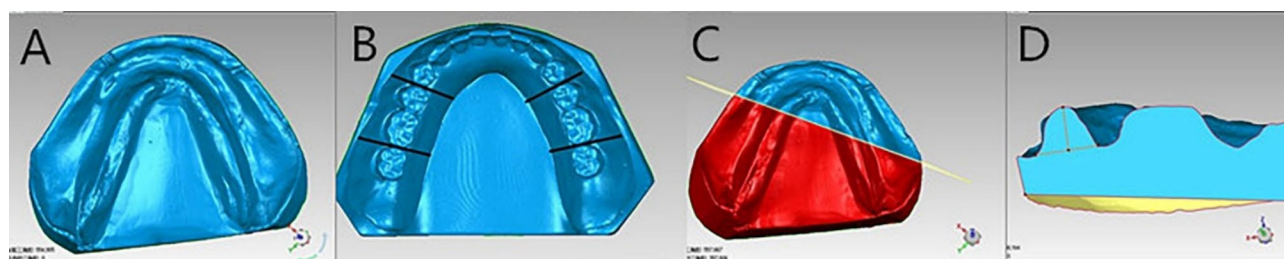


Fig. 4 **A.** Data imported into the software. **B.** The selection of alveolar ridge cross-sectional positions in the premolar and molar areas. **C.** Trim with plane. **D.** The width and height of the mandibular alveolar ridge

Study outcomes

In this study, the OCA, OCN, and bite force were the primary outcomes, the wear of artificial teeth was the secondary outcome.

Measurements of mandibular alveolar ridge

The height and width of the mandibular alveolar ridge in the premolar and molar areas were measured in this study because the morphology of the alveolar ridge significantly influences the stability of complete dentures which could consequently affect some of the outcomes investigated. The width and height of the mandibular alveolar ridge were measured using the software (Geomagic Studio 2013, Raindrop); the mandibular master casts were scanned using a scanner (X5; Sirona) and subsequently imported into the Geomagic Studio 2013 (Fig. 4A). The premolar (between the first and second mandibular premolars) and molar areas (between the first and second mandibular molars) (Fig. 4B) were selected based on the anatomical features and position of the artificial tooth. The trim with plane (Fig. 4C) and distance functions in the software were used to measure the height and width of the left and right alveolar ridges in the premolar and molar areas (Fig. 4D). All the data were measured three times, and the average was recorded [24].

OCA and OCN assessment

OCA and OCN were measured using the silicone techniques (Blue Silicone, GC). The chair was adjusted to make the participants sit upright with the occlusal plane parallel to the floor. The mandibular occlusal surface was covered by silicone. After ensuring the correct occlusal position, the participants were instructed to maintain the maximum bite force for 1 min. After solidifying, silicone rubber was removed, cleaned with running water, and surplus rubber was clipped. Occlusal contact analyzer software (BiteEye, GC) was launched, “Patient selection/Editing” was selected to create a participant profile, and “Photography & Analysis” was pressed. The occlusal silicone rubber was inserted into the supporting device (Fig. 5A). The camera icon was clicked, and Lv2 (100 μ m) was picked (Fig. 5B). The software was programmed to calculate and generate the results automatically [6, 9, 11].

Maximum bite force evaluation

The chair was adjusted to make the participants sit upright with the occlusal plane parallel to the floor. Dental Prescale II (DP II, GC, Japan) that suited the participants' arch was selected and rotated into their mouth. DP II's anterior triangle apex was confirmed to be in line with the midline of the anterior teeth, and each tooth was equally capped. The participants were instructed to apply the maximum force for 1 min. DP II was placed

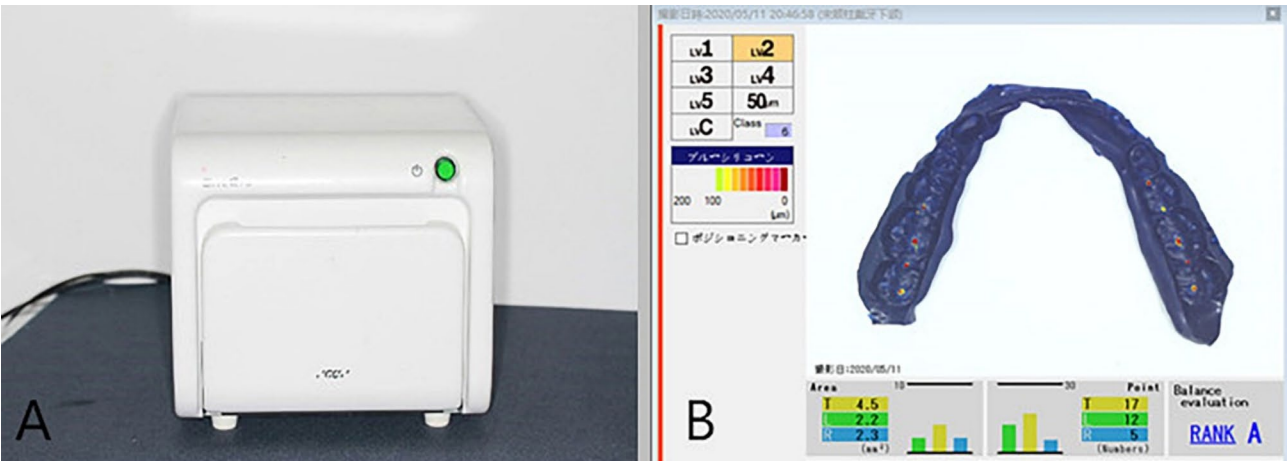


Fig. 5 A. Bite eye. B. OCA and OCN measurements

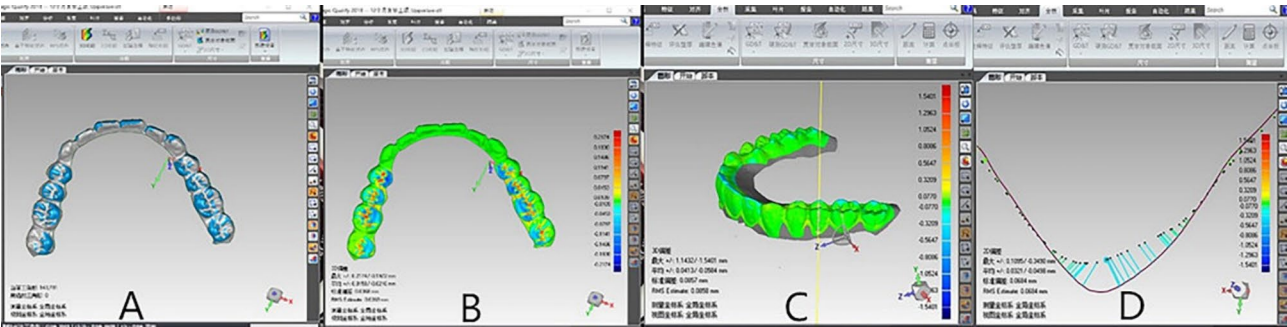


Fig. 6 A. Best fit alignment. B. Three-dimensional analysis. C. Creates a survey section. D. Comparative result

Table 1 The distribution of edentulism among the three groups

	Cases (n)	Age	Male	Female	ClassII	Class III
Group I	10	68	5	5	4	6
Group II	10	67	6	4	4	6
Group III	10	66	5	5	5	5

into the scanner, and the software (Bite Force Analyzer, GC) was launched. “Select patient” was selected to create the record file, and the “Scan” button was clicked to scan. Data was read and saved using the software [8, 10].

The wear of artificial teeth measurements

The software was used to measure artificial teeth wear (Geomagic Qualify 2013, Raindrop). Maxillary and mandibular dentures were scanned at 2-week and 12-month follow-up visits using the Sirona X5 scanner. Following this, the data were imported into the program. Two-week data was set as “Reference” and 12-month data was the “Test”. The severe wear points of the buccal cusp, lingual cusp, and occlusal fossa were identified using best fit alignment (Fig. 6A) and 3D analysis (Fig. 6B). Vertical cross-sections were then created at these points (Fig. 6C), and the vertical loss of artificial teeth was assessed, the 2D comparison function was used to measure the linear

distance (Fig. 6D). Each portion was measured thrice in the same manner and the average was calculated [25].

Statistical analysis

One-way analysis of variance was used to analyze the OCA, OCN, bite force, wear of artificial teeth, as well as the premolar and molar height and width of mandibular alveolar ridges after the normal distribution was confirmed with the Shapiro-Wilk test. All pairwise statistical analyses were performed with the LSD test. All statistical analyses were performed using SPSS 20.0 (IBM Corp., USA), with a significance level of <0.05.

Results

The study included 30 participants, as shown in Table 1. There were no differences in age and sex among the three groups. Figure 7 illustrates the study flow diagram and the reason for attrition.

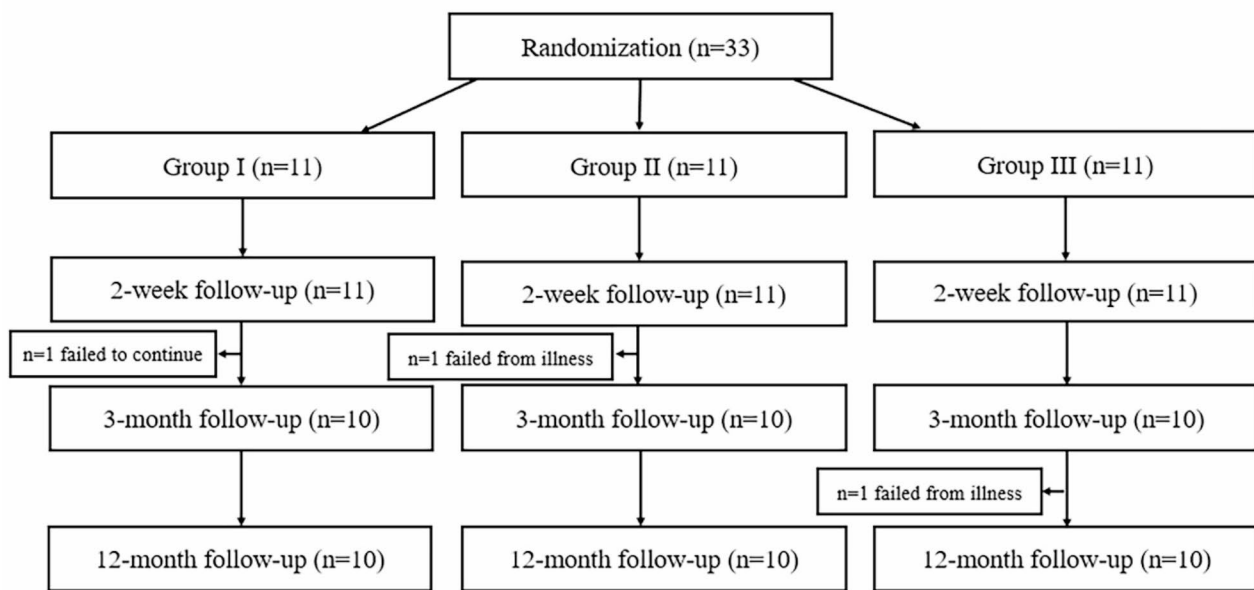


Fig. 7 The flow diagram of the study

Table 2 Results of the one-way analysis of variance (ANOVA) test for mandibular ridge height and width (mm)

	Premolar Area		Molar Area	
	Height Mean $\pm$ SD	Width Mean $\pm$ SD	Height Mean $\pm$ SD	Width Mean $\pm$ SD
Group I	6.71 $\pm$ 1.91	12.38 $\pm$ 2.66	6.56 $\pm$ 1.97	13.70 $\pm$ 2.62
Group II	5.47 $\pm$ 3.02	11.19 $\pm$ 2.57	5.65 $\pm$ 3.10	12.34 $\pm$ 3.32
Group III	7.11 $\pm$ 2.42	12.43 $\pm$ 1.70	6.99 $\pm$ 2.17	13.81 $\pm$ 1.10
F	1.169	0.857	0.759	1.009
p	0.326	0.436	0.478	0.378

The three groups did not differ significantly in height and width of the mandibular alveolar premolar and molar ridge areas ($P > 0.05$) (Table 2). The potential confounding effect of the variations in alveolar ridge dimensions on the study outcomes could be dismissed.

The results exhibited that the OCA of group III was significantly higher than that of groups I and II at the 2-week follow-up visits, whereas the OCA of group I was significantly higher than groups II and III at the 3-month and 12-month follow-up visits ($P < 0.05$) (Fig. 8A). The OCN of the three groups was significantly different at the three follow-up visits ($P < 0.05$) (Fig. 8B).

The bite force of groups I and II was significantly lower than that of group III at the three follow-up visits ($P < 0.05$). No significant difference was observed in the bite force between groups I and II at the 2-week, 3-month, and 12-month follow-up visits ($P > 0.05$) (Fig. 8C).

During the 12-month follow-up visits, statistically significant differences were observed in the average wear

among the three groups ($P < 0.05$) (Fig. 8D). Figure 9 illustrates the posterior artificial teeth wear.

Discussion

The primary study findings were significant differences in the OCA, OCN, bite force, and the wear of artificial teeth among the three groups. Consequently, the null hypothesis was rejected.

Masticatory performance determines the ability to effectively digest food, serving as an objective measure of oral function. Several parameters, such as bite force, occlusal condition, and OCA, affect this performance [6, 9, 26]. A positive correlation exists between occlusal area and masticatory performance [6, 27, 28]. Moreover, the OCA increased with the bite force applied to the impression material during the verification of interocclusal records [6, 29]. The participants were instructed to maintain their maximal bite force for 1 min to minimize fluctuations in the OCA caused by bite force [9, 29].

This study demonstrated that participants with LO exhibited significantly lower OCA and OCN compared to those with BBO at the 2-week follow-up visit. This difference may be attributed to the fact that, in LO, only the maxillary lingual cusps contacted the mandibular central fossae [3, 30]. No significant difference in OCA was observed between groups I and II, while a notable difference was observed in OCN. This discrepancy could be because of the differences in the cusp inclination and occlusal surface morphology of the artificial teeth [30].

This study recognized a correlation between OCA and OCN and the wear of artificial teeth, indicating that the latter was a continuous process [20, 21]. Both OCA and

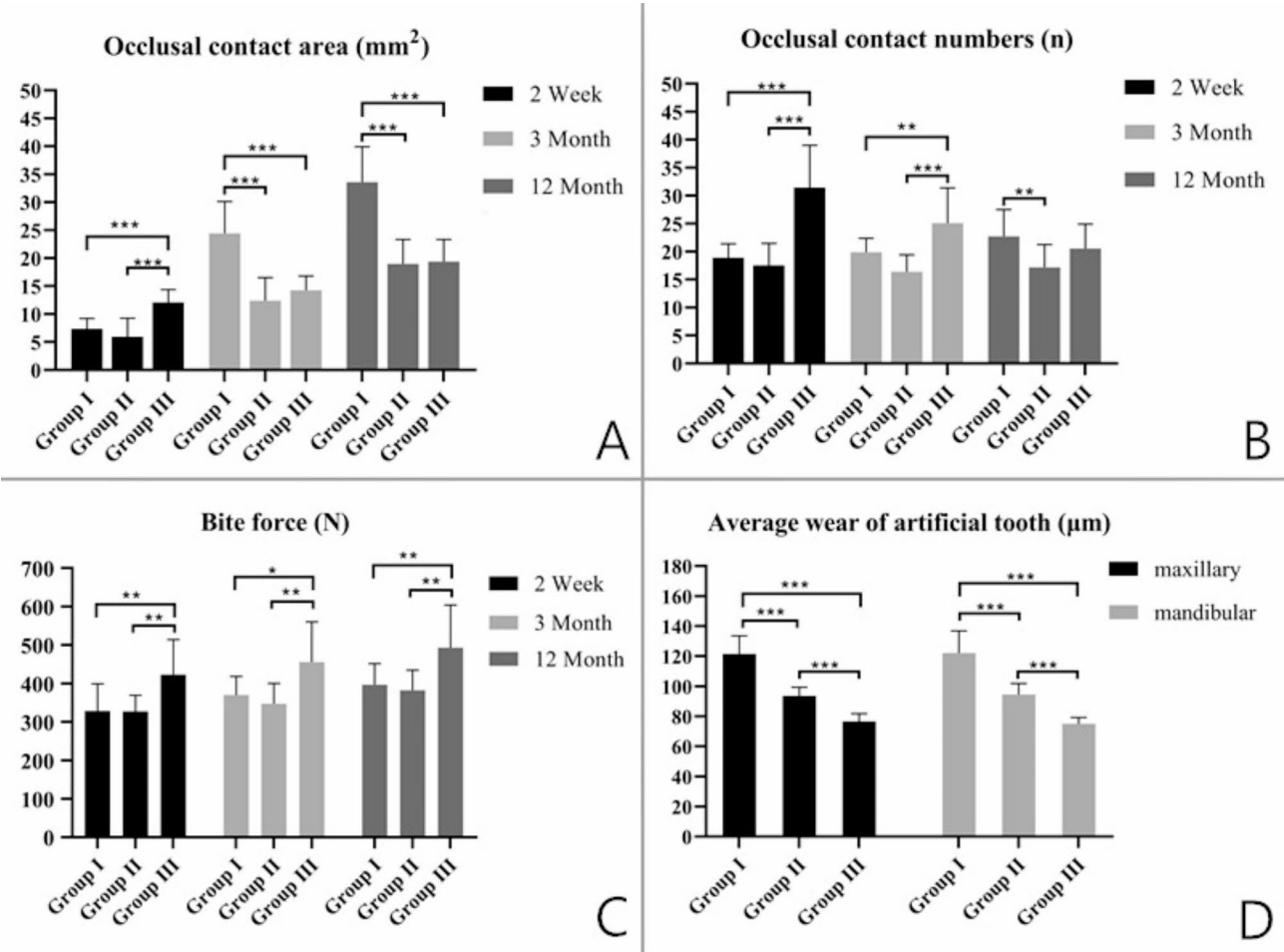


Fig. 8 **A** shows the OCA. **B** illustrates the OCN. **C** demonstrates the bite force of the three groups. **D** illustrates the average wear of artificial teeth. * Indicates significant differences in the comparison of one-way ANOVA tests between groups. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

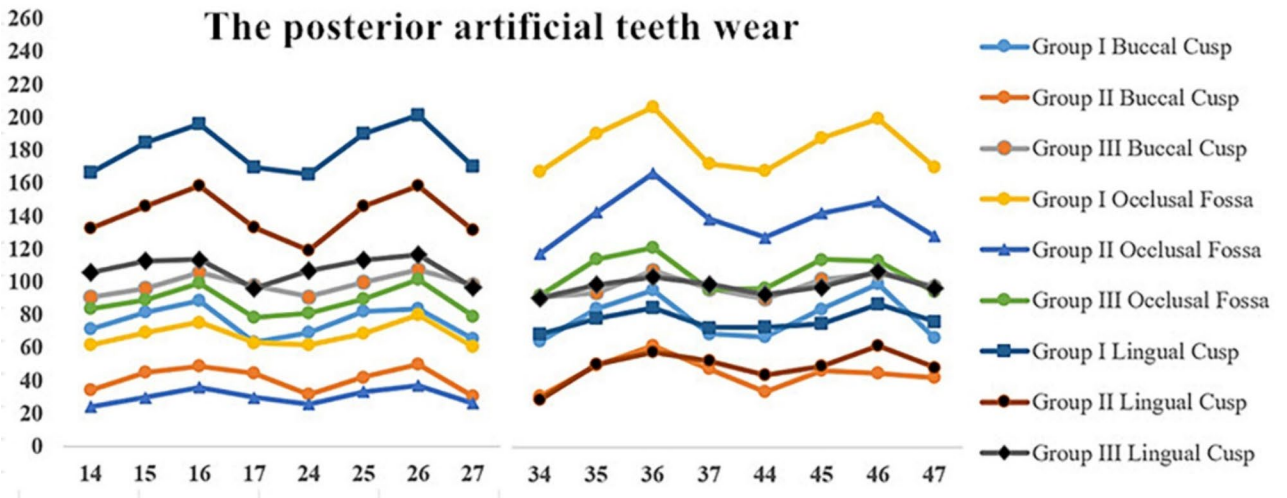


Fig. 9 The posterior artificial teeth wear

OCN exhibited notable changes across all three groups at the 3-month follow-up visit. OCA demonstrated an increase, while OCN experienced a decrease. This could be attributed to the occlusal contacts possibly undergoing a smoothing process after 3 months of wear, leading to the formation of larger contact points as the smaller surrounding surfaces were abraded. This finding is in accordance with the earlier research demonstrating that the wear might be clinically detected in a short time [21]. The OCA in the complete dentures with LO and BBO exhibited further increase at the 12-month follow-up visit. The OCA increased because the transition from point contact to plane contact was caused by the wear of artificial teeth. The observed increase in the OCN in the LO groups could be attributed to the contact between the buccal cusps of the premolars and molars with the opposing jaw.

The bite force, assessed in oral biomechanics, is the most significant indicator of masticatory system health [28, 31, 32]. The maximum bite force indicates masticatory ability, providing objective data for assessing the occlusal system, joints, and masticatory muscles [33]. Morokuma et al. [34] found that after the complete denture treatment, the bite force increased to 277.2 N. Patil et al. [35] observed that the mean bite force in complete denture wearers was 404.9, 229.5, and 321.5 N for men, women, and the total group, respectively. The results of this study on bite force were similar to those of Patil et al.; however, higher than Morokuma et al. since the occlusal force was affected by age, sex, health status, shape of the face, masticatory muscle strength, and tooth contact area, impacting the amount of bite force [36, 37]. This study demonstrated that the bite force increased across all three groups during the 12-month observation period, which aligns with the findings of Shala et al. [32]. This could be associated with the participants' need to gradually adjust to the new complete denture, including adaptation to the new occlusal scheme, nerve and muscle adaptation, potential occlusal interference wear, and psychological adaptation [3, 38]. The adjustment time of participants varies and may take weeks or even months [3, 23]. The study also found that at the three follow-up visits, LO had a lower bite force than BBO. This might be because LO reduced masticatory muscle activation and mastication duration more than BBO [38].

Acrylic resin teeth were introduced in the early 1930s and widely used for denture fabrication [39]. The wear behavior of acrylic artificial teeth in the oral cavity was primarily influenced by the wear resistance and was determined by the materials used in tooth fabrication [14, 17, 39]. Modified acrylic resin teeth contained cross-linked or interpenetrating network structures and appended microfillers or nanofillers; thus, improving their physical properties [12, 39].

Stober et al. [17] found that the maximum vertical loss of composite resin denture teeth at 6, 12, and 24 months was 92, 146, and 226 μm , respectively. Schmid-Schwab et al. [21] measured median vertical loss for 121–221 μm posterior cross-linked denture teeth, after 1 year of clinical use. Heintze et al. [20] observed wear of 250 μm for DCL denture teeth after 2 years of clinical service. In this study, the wear of groups I, II, and III after a year of clinical service was 121.75, 93.96, and 75.78 μm , respectively. The study findings were slightly lower than those of previous studies, perhaps because of individual differences and dietary habits [21, 40]. This study also found that the bite force of group III was larger than that of groups I and II; however, the wear loss was lower than that of groups I and II, indicating that the artificial tooth wear resistance was strongly connected to its internal structures and fillers [12, 39], the bite force was a secondary influence.

Consistent wear was observed in the maxillary cusps, occlusal fossae, mandibular buccal cusps, and central fossae of BBO. Conversely, wear primarily occurred in LO's maxillary lingual cusps and mandibular occlusal fossae. The observed differences in wear patterns could be ascribed to the distinctive occlusal characteristics of the two schemes. Furthermore, differences in the wear distribution of artificial posterior teeth were observed between the maxillary and mandibular dentures in groups I and II. The lingual inclined planes of the maxillary buccal cusps and buccal and lingual inclined planes of the mandibular buccal cusps demonstrated varying levels of wear. This could be because of the friction between foods and artificial teeth, as well as the direct contact of the artificial teeth [14]. The lateral movement caused wear on the lingual and buccal inclined planes of the mandibular buccal cusps [14]. This study also showed that the wear of the second premolar and first molar was higher than that of the first premolar and second molar, thereby corroborating the findings of the study by Heintze et al. [20]. This could be because the center of mastication was situated at this location, and the bite force was greater than that of the other two regions [9, 32].

This study has few limitations. The sample size was small, and only the alveolar ridge classification was considered. Although statistically significant differences in the occlusal parameters and wear of artificial teeth were found, further studies should be conducted with larger sample sizes and with a stratified randomized design, based on age, sex, dietary habits, face morphology, and masticatory muscle status. This would help in the evaluation of the differences between lingualized and bilateral balanced occlusal complete dentures more efficiently.

Conclusions

As the duration of use increases, the occlusal contact area of complete dentures increases, regardless of the occlusal schemes. The bite force of lingualized occlusion is lower than that of bilateral balanced occlusion. In lingualized occlusion, wear primarily occurred on the maxillary lingual cusps and mandibular occlusal fossae of artificial teeth. However, contact between the maxillary buccal cusps and mandibular posterior teeth might occur after some time.

Abbreviations

BBO	Bilateral balanced occlusion
DCL	Double cross-linked
DP II	Dental Prescale II
IPN	Interpenetrating polymer networks
LO	Lingualized occlusion
OCA	Occlusal contact area
OCN	Occlusal contact number

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12903-024-05144-2>.

Supplementary Material 1

Supplementary Material 2

Acknowledgements

This study was supported by grants from the Natural Science Foundation of Liaoning Province, China [Grant numbers 2019-MS-074].

Author contributions

Zhiqiang Wang: Participate in the design and development of research protocols, implementation of the research process, collation of data, statistical analysis, drafting and revision of papers. Yan Su: Participate in data counting, checking, analyzing, and revising papers. Jianmin Wang: fabricate the complete dentures. Yang Liu: Participate in the design and development of the research program and guide to carry out the implementation. Wenzhong Xing: Participate in proposing research topics, designing and developing research proposals, mentoring support, researching and organizing literature, revising papers.

Funding

This research received no external funding.

Data availability

Data that support the findings of this study are available from the corresponding author, upon reasonable request.

Declarations

Ethics approval and consent to participate

The study was carried out in compliance with the Helsinki Declaration, approved by the Medical Ethics Committee of Dalian Stomatological Hospital (Ethics Review Batch No. DLKQLL2019015, 28/11/2019) and retrospectively registered in the Chinese Clinical Trial Registry (No. ChiCTR2300073420, 11/07/2023). Informed consent was obtained from all subjects involved in the study.

This study adheres to CONSORT guidelines.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 21 June 2024 / Accepted: 4 November 2024

Published online: 19 November 2024

References

- Limpuangthip N, Somkotra T, Arksornnukit M. Modified retention and stability criteria for complete denture wearers: a risk assessment tool for impaired masticatory ability and oral health-related quality of life. *J Prosthet Dent*. 2018;120(1):43–9.
- Goldstein G, Kapadia Y, Campbell S. Complete denture occlusion: best evidence consensus statement. *J Prosthodont*. 2021;30(5):72–7.
- Lemos C, Verri FR, Gomes J, Júnior JS, Moraes S, Pellizzer EP. Bilateral balanced occlusion compared to other occlusal schemes in complete dentures: a systematic review. *J Oral Rehabil*. 2018;45(4):344–54.
- Kawai Y, Ikeguchi N, Suzuki A, Kuwashima A, Sakamoto R, Matsumaru Y, Kimoto S, Iijima M, Feine JS. A double blind randomized clinical trial comparing lingualized and fully bilateral balanced posterior occlusion for conventional complete dentures. *J Prosthodont Res*. 2017;61(2):113–22.
- Matsumaru Y. Influence of mandibular residual ridge resorption on objective masticatory measures of lingualized and fully bilateral balanced denture articulation. *J Prosthodont Res*. 2010;54(3):112–8.
- Lee HS, Ko KH, Huh YH, Cho LR, Park CJ. Correlation between occlusal contact area at various levels of interocclusal thicknesses and masticatory performance. *J Oral Rehabil*. 2022;49(5):522–8.
- Terauchi R, Arai K, Tanaka M, Kawazoe T, Baba S. Effect of difference in occlusal contact area of mandibular free-end edentulous area implants on periodontal mechanosensitive threshold of adjacent premolars. *Springerplus*. 2015;4:703.
- Zhao Z, Wang Q, Li J, Zhou M, Tang K, Chen J, Wang F. Construction of a novel digital method for quantitative analysis of occlusal contact and force. *BMC Oral Health*. 2023;23(1):190.
- Imamura Y, Sato Y, Kitagawa N, Uchida K, Osawa T, Omori M, Okada Y. Influence of occlusal loading force on occlusal contacts in natural dentition. *J Prosthodont Res*. 2015;59(2):113–20.
- Shiga H, Komino M, Uesugi H, Sano M, Yokoyama M, Nakajima K, Ishikawa A. Comparison of two dental prescale systems used for the measurement of occlusal force. *Odontology*. 2020;108(4):676–80.
- Horie T, Kanazawa M, Komagamine Y, Hama Y, Minakuchi S. Association between near occlusal contact areas and mixing ability. *J Oral Rehabil*. 2014;41(11):829–35.
- Stober T, Henninger M, Schmitter M, Pritsch M, Rammelsberg P. Three-body wear of resin denture teeth with and without nanofillers. *J Prosthet Dent*. 2010;103(2):108–17.
- Gad MM, Alalawi H, Akhtar S, Al-Ghamdi R, Alghamdi R, Al-Jefri A, Al-Qarni FD. Strength and wear behavior of three-dimensional printed and prefabricated denture teeth: an in vitro comparative analysis. *Eur J Dent*. 2023;17(4):1248–56.
- Saadi ASA, El-Damanhoury HM, Khalifa N. 2D and 3D wear analysis of 3D printed and prefabricated artificial teeth. *Int Dent J*. 2023;73(1):87–92.
- Suzuki S. In vitro wear of nano-composite denture teeth. *J Prosthodont*. 2004;13(4):238–43.
- Heintze SD, Zellweger G, Grunert I, Munoz-Viveros CA, Hagenbuch K. Laboratory methods for evaluating the wear of denture teeth and their correlation with clinical results. *Dent Mater*. 2012;28(3):261–72.
- Stober T, Geiger A, Rues S, Dreyhaupt J, Rammelsberg P, Ohlmann B. Factors affecting wear of composite resin denture teeth—24-month results from a clinical study. *Clin Oral Investig*. 2012;16(2):413–20.
- Heintze SD, Rousson V, Stober T. Patient- and therapy-related factors on the wear of denture teeth—results of a clinical trial. *Dent Mater*. 2015;31(3):302–7.
- Pham DM, Gonzalez MD, Ontiveros JC, Kasper FK, Frey GN, Belles DM. Wear resistance of 3D printed and prefabricated denture teeth opposing zirconia. *J Prosthodont*. 2021;30(9):804–10.
- Heintze SD, Zellweger G, Sbracego S, Rousson V, Munoz-Viveros C, Stober T. Wear of two denture teeth materials in vivo—2-year results. *Dent Mater*. 2013;29(9):e191–204.
- Schmid-Schwab M, Rousson V, Vornwagner K, Heintze SD. Wear of two artificial tooth materials in vivo: a 12-month pilot study. *J Prosthet Dent*. 2009;102(2):104–14.
- Zhang P, Xu J. Study on retention and stability of linear occlusal complete dentures. *Zhonghua Kou Qiang Yi Xue Za Zhi*. 2003;38(1):46–8.

23. Limpuangthip N, Tumrasvin W, Pinyowattanasilp T, Kulpiyavaja P. Evaluation of dynamic impression lining material effects on masticatory function and the oral health-related quality of life of complete denture wearers: a six-month randomized controlled trial. *J Prosthodont Res.* 2023;67(2):214–21.
24. Xia Y, Ma C, Chen J, Witter DJ, Zhang Q, Creugers NHJ. Mandibular residual ridge morphology in relation to complete dentures and implant overdentures-part I: predictors for perceived conventional denture stability. *Clin Implant Dent Relat Res.* 2021;23(1):131–9.
25. Kühne C, Lohbauer U, Raith S, Reich S. Measurement of tooth wear by means of digital impressions: an in-vitro evaluation of three intraoral scanning systems. *Appl Sci.* 2021;11(11):5161.
26. Korda B, Amlang A, Hugger A, Behrendt C, Ruge S. Number and localization of occlusal contact areas on natural posterior teeth without dental findings - evaluations of the regional baseline study (SHIP-1) with the Greifswald Digital Analyzing System (GEDAS). *Int J Comput Dent.* 2022;25(1):47–56.
27. Owens S, Buschang PH, Throckmorton GS, Palmer L, English J. Masticatory performance and areas of occlusal contact and near contact in subjects with normal occlusion and malocclusion. *Am J Orthod Dentofac Orthop.* 2002;121(6):602–9.
28. Lepley CR, Throckmorton GS, Ceen RF, Buschang PH. Relative contributions of occlusion, maximum bite force, and chewing cycle kinematics to masticatory performance. *Am J Orthod Dentofac Orthop.* 2011;139(5):606–13.
29. Hidaka O, Iwasaki M, Saito M, Morimoto T. Influence of clenching intensity on bite force balance, occlusal contact area, and average bite pressure. *J Dent Res.* 1999;78(7):1336–44.
30. Engelmeier RL, Phoenix RD. The development of lingualized occlusion. *J Prosthodont.* 2019;28(1):e118–31.
31. Iwase M, Ohashi M, Tachibana H, Toyoshima T, Nagumo M. Bite force, occlusal contact area and masticatory efficiency before and after orthognathic surgical correction of mandibular prognathism. *Int J Oral Maxillofac Surg.* 2006;35(12):1102–7.
32. Shala K, Tmava-Dragushe A, Dula L, Pustina-Krasniqi T, Bicaj T, Ahmedi E, Lila Z. Evaluation of maximum bite force in patients with complete dentures. *Open Access Maced J Med Sci.* 2018;6(3):559–63.
33. Sonogo MV, Goiato MC, Dos Santos DM. Electromyography evaluation of masseter and temporalis, bite force, and quality of life in elderly patients during the adaptation of mandibular implant-supported overdentures. *Clin Oral Implants Res.* 2017;28(10):e169–74.
34. Morokuma M, Yoneyama Y, Matsuda R, Hosoi T, Ohkubo C. Influence of occlusal force on electroencephalograms in edentulous patients. *J Prosthodont.* 2015;24(7):532–7.
35. Patil SR, Maragathavalli G, Sghaireen MG, Al-Zoubi IA, Vargheese S, Alam MK. Maximum bite force in complete denture wearing patients: a preliminary study. *Int Med Journal: IMJ.* 2019;26(3):220–2.
36. Thongudomporn U, Chongsuvivatwong V, Geater AF. The effect of maximum bite force on alveolar bone morphology. *Orthod Craniofac Res.* 2009;12(1):1–8.
37. Koc D, Dogan A, Bek B. Bite force and influential factors on bite force measurements: a literature review. *Eur J Dent.* 2010;4(2):223–32.
38. Deniz DA, Kulak Ozkan Y. The influence of occlusion on masticatory performance and satisfaction in complete denture wearers. *J Oral Rehabil.* 2013;40(2):91–8.
39. Kamonwanon P, Yodmongkol S, Chantarachindawong R, Thaweeboon S, Thaweeboon B, Srihirin T. Wear resistance of a modified polymethyl methacrylate artificial tooth compared to five commercially available artificial tooth materials. *J Prosthet Dent.* 2015;114(2):286–92.
40. Zeng J, Sato Y, Ohkubo C, Hosoi T. In vitro wear resistance of three types of composite resin denture teeth. *J Prosthet Dent.* 2005;94(5):453–7.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.